

Assignment

Python-Fundamentals-of-python-language

Session-1

Q-5). Explain the difference between how Python executes code versus how C or Java executes code, using your own words and a simple example for each.

Answer. Python, C, and Java are programming languages, but they execute programs in different ways.

Python:-

Python is generally called an interpreted language. When we run a Python program, Python converts the source code into bytecode and executes it using the Python interpreter/runtime.

Example:

```
print("Hello World")
```

When we run this program, Python executes the print() statement and displays:

Hello World

C:-

C is a compiled language. First, the C source code is compiled by a compiler into machine code. After compilation, the generated executable program can be run directly by the computer.

Example:

```
#include <stdio.h>
```

```
int main()  
{  
    printf("Hello World");  
    return 0;  
}
```

The C compiler converts this program into machine code, and then the executable file is run to display:

Hello World

Java:-

Java is a **compiled and platform-independent language**. First, the Java source code is compiled by the Java compiler into **bytecode**. This bytecode is stored in a .class file and then executed by the **Java Virtual Machine (JVM)**.

Example:

```
class Hello
{
    public static void main(String[] args)
    {
        System.out.println("Hello World");
    }
}
```

The Java compiler converts this program into **bytecode**, and then the JVM executes the bytecode to display:

Hello World